# AutoDSC: Automated Algorithm Selection and Tuning for Non-Stationary Data Stream Clustering

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***Abstract*—We introduce *AutoDSC* (Automated Data Stream Clustering), a novel AutoML framework that automatically selects and tunes clustering algorithms for evolving data streams. This matters because in IoT, network monitoring, and sensor applications, the data distribution changes continuously, so no fixed algorithm or configuration stays optimal, yet clustering lacks a universal objective to optimise, which is why AutoML has transformed supervised learning, yet its extension to unsupervised streams remains largely unexplored. *AutoDSC* achieves fully autonomous adaptivity by continuously searching the joint algorithm and hyperparameter space using population-guided stochastic search optimisation and selecting the best configuration at each window stage using a multi-objective quality measure, without ground-truth labels. We benchmark AutoDSC against state-of-the-art stream-clustering and AutoML methods on ten representative real-world and synthetic data streams. AutoDSC attains the best overall rank in clustering quality among all compared methods while remaining computationally efficient.**

***Index Terms*—data stream clustering, AutoML, algorithm selection, non-stationary, unsupervised learning, multi-objective optimisation**

## I. Introduction

Data stream clustering is a fundamental operation in a wide range of real-world applications, from IoT sensor networks and network intrusion detection to social media analysis and financial monitoring [1], [2]. Unlike batch clustering, stream clustering must process data continuously, in a single pass, while adapting to a distribution that changes over time.

Numerous stream clustering algorithms have been proposed [3]–[6], each effective under specific conditions. *However, the right algorithm and its hyperparameters must themselves change as the stream evolves — no single configuration stays optimal throughout.* Existing automated solutions either rely on a single evaluation metric [7], require computationally intensive offline tuning [8], or reconfigure structural components without jointly searching the algorithm–hyperparameter space [9], leaving a clear gap for a fully autonomous, label-free AutoML approach to stream clustering.

In this paper, we introduce **AutoDSC** (Automated Data Stream Clustering), a novel AutoML framework that automatically selects and tunes a clustering algorithm online, in a single pass and without labels, as the data distribution evolves. AutoDSC maintains a population of candidate configurations, continuously searches the joint algorithm-and-hyperparameter space using **Population-Guided Stochastic Search Optimisation (PGSSO)**, and selects the best configuration at each window stage using a multi-objective, label-free quality measure. We evaluate AutoDSC against five baselines and make the following key contributions.

- We formalise online stream clustering as a combined algorithm selection and hyperparameter optimisation (CASH) problem that must be solved continuously, window stage by window stage, as the data distribution evolves, and present AutoDSC, a single-pass AutoML framework that solves it autonomously without manual tuning or labels.
- AutoDSC introduces **PGSSO**, which maintains a recency-weighted elite archive and an adaptively controlled exploration rate so that promising configurations are found quickly and the population recovers implicitly when the distribution changes.
- AutoDSC scores each configuration with a normalised multi-objective criterion combining three complementary internal cluster-validity indices, yielding robust selection where the single-metric criteria of prior systems fail.
- We provide a comprehensive empirical evaluation across ten real-world and synthetic streams, including statistical significance testing, ablation, sensitivity, search-overhead decomposition, and dimensionality-scalability analyses, demonstrating that AutoDSC consistently ranks first in clustering quality and is the most computationally efficient of the AutoML stream clusterers.

The rest of the paper is organised as follows. Section II introduces the background of data stream clustering and AutoML. Section III formalises the problem. Section IV presents the AutoDSC framework. Section V describes the experimental setup. Section VI presents empirical evaluation and analysis. Section VII reviews prior work, and Section IX concludes the paper.

## II. Background

We review the data stream clustering setting, how clustering quality is measured online, and the automated machine learning concepts that AutoDSC builds on.

### A. Data Stream Clustering

A data stream is an unbounded, high-velocity sequence of instances that a learner must process online rather than as a stored dataset. Following Bifet et al. [10], a stream learner should satisfy five requirements: (**R1**) process one instance at a time and inspect it at most once; (**R2**) use a bounded amount of time per instance; (**R3**) use a bounded amount of memory; (**R4**) be ready to produce an answer, such as a prediction, clustering, or set of patterns, at any time; and (**R5**) adapt to temporal changes in the data. These constraints rule out the multi-pass, full-data access that batch clustering and offline

AutoML assume. Because the data-generating process evolves, streams also exhibit concept drift, meaning the distribution of the data, and for clustering the number, shape, and location of the clusters, changes over time, whether abruptly, gradually, or incrementally [10], [11]. A method that fixes its model or configuration in advance therefore degrades as the stream moves away from the regime it was tuned for, which is exactly what makes requirement R5 hard.

Classical stream clustering algorithms meet these requirements and fall into a few families distinguished by how they summarize the stream. The line begins with BIRCH [12], whose clustering-feature tree first showed that a single-scan, bounded-memory summary could cluster very large datasets and seeded the micro-cluster summaries used by later stream methods. Micro-cluster summarization, pioneered by CluStream [3], maintains pyramidal time-frame statistics for scalable analysis; density-based methods such as DenStream [4] and DBSTREAM [5] discover arbitrarily shaped clusters through core, potential, and outlier micro-clusters and a shared-density graph; and k-means-based methods maintain centroids online, where mini-batch optimisation enables web-scale clustering at low latency [13] and StreamKMeans [6] replaces the k-medians step of stream local search [14] with incremental k-means. These algorithms are effective once configured, but each fixes its algorithm and hyperparameters in advance, which motivates automating their selection. Quality is measured online with the interleaved test-then-train, or prequential, protocol [10], in which each instance is used to test the model before it updates on it. As labels are unavailable during the stream, configuration choices rely on internal cluster validity indices (CVIs) such as the Silhouette, Davies–Bouldin, and Calinski–Harabasz scores, and since no single CVI is reliable across cluster shapes, densities, and counts, combining complementary indices is more robust than any one alone [15], [16]; for label-based scoring, Kremer et al. [17] introduced the Cluster Mapping Measure (CMM), which unlike classical external indices accounts for emerging, splitting, and moving clusters, so we evaluate with CMM alongside the chance-corrected Adjusted Rand Index (Section V).

### *B. Automated Machine Learning*

Automated Machine Learning removes the manual burden of algorithm selection and hyperparameter optimisation, formalised as the CASH problem [18]–[20]. It is mature for batch supervised learning, whereas clustering lacks a universally applicable objective and streaming imposes single-pass, bounded-memory, and drifting constraints, so existing automated systems address at most one of these (surveyed in Section VII). AutoDSC casts stream clustering as an online joint algorithm–hyperparameter search, which we formalise next.

## III. Problem Formulation

A data stream is an unbounded sequence $\mathcal{S} = \{x_1, x_2, \ldots, x_t, \ldots\}$ in which each instance $x_t \in \mathbb{R}^d$ arrives one at a time and must be processed in a single pass under bounded memory. We consume the stream in consecutive windows of fixed size $w$, writing $W$ for the window currently being processed, and the data-generating distribution may change from one window to the next (concept drift), so the best way to cluster the data is itself time-varying. Given a window $W$, a clustering algorithm groups its points by similarity without access to labels. Centroid-based methods, for instance, seek centroids that minimize the average distance from each point to its nearest centroid, while density- and micro-cluster-based methods optimise related criteria. No single algorithm or objective suits all data, so the appropriate choice varies across datasets and, under drift, across time. Because ground-truth labels are unavailable online, the quality of a clustering produced on $W$ must be judged by an internal cluster-validity score $q(\cdot) \in [0, 1]$ (higher is better) computed from the data and the assignments alone, whereas external indices such as ARI and CMM use labels and are reserved for offline evaluation (Section V).

Let $\mathcal{A}$ be a portfolio of stream-clustering algorithms and $\Theta = \bigcup_{a \in \mathcal{A}} (\{a\} \times \Phi_a)$ the joint space of configurations $\theta = (a, \phi)$ that pair an algorithm $a$ with hyperparameters $\phi \in \Phi_a$. Running $\theta$ on the current window yields a clustering with label-free quality $q(\theta; W)$. The CASH problem for a stream is to track, at every window, the configuration

$$\theta^\star = \arg\max_{\theta \in \Theta} \; q(\theta; W), \tag{1}$$

using only the data seen so far and bounded memory. As the distribution drifts, $\theta^\star$ changes over time, so a solver must continually re-evaluate and re-select configurations rather than commit to one. AutoDSC approximates this online optimum with a population-guided search and a label-free multi-objective instantiation of $q$, described next.

## IV. Automated Data Stream Clustering

We introduce Automated Data Stream Clustering (AutoDSC), a novel AutoML framework tailored for non-stationary data stream clustering. AutoDSC processes data in fixed-size windows. At the end of each window, AutoDSC evaluates clustering configurations using multiple normalised validation indices. The framework maintains a recency-weighted elite archive of high-performing configurations to guide future parameter exploration via PGSSO. It adaptively balances exploration and exploitation through dynamic rate adjustment based on performance feedback. Adaptation to distributional change is implicit, since, in each window, the population is re-evaluated and the incumbent re-selected, and a drop in clustering quality automatically raises the exploration rate so new configurations are sought for the new regime. Figure 1 illustrates the high-level workflow of AutoDSC, while Algorithm 1 details the complete operational steps.

To solve the online CASH problem of Section III, AutoDSC maintains a configuration population $\mathcal{A}_t \subseteq \Theta$ of clustering algorithms with different hyperparameter settings, fills the current window $W_t$ as instances arrive, and serves real-time predictions from the best-performing (incumbent) configuration. As each new data point arrives, every configuration

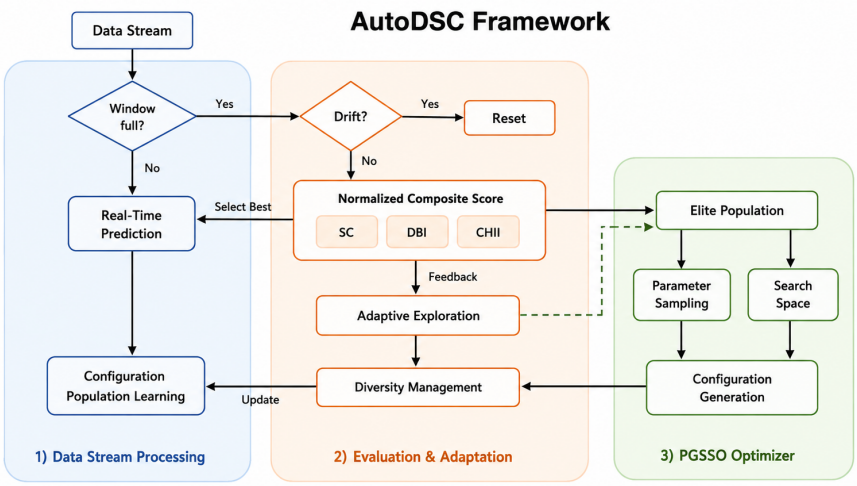



Fig. 1. Overview of AutoDSC. Each window the population of clustering configurations is evaluated by the multi-objective criterion, the incumbent is re-selected, and PGSSO proposes new configurations from the recency-weighted elite archive under an adaptive exploration rate.

in the population updates incrementally on it with a single LEARNONE step, retaining the per-point cluster assignment for later scoring. When the window reaches capacity, the framework initiates evaluation and optimisation.

When the window fills, AutoDSC evaluates clustering quality using three normalised clustering validation indices, namely the Normalised Silhouette Coefficient $\text{NSC} = (1 + \text{SC})/2$, the Normalised Davies-Bouldin Index $\text{NDBI} = 1 - \tanh(\text{DBI})$, and the Normalised Calinski-Harabasz Index $\text{NCHI} = \tanh(\log(1 + \text{CHI})/10)$, ensuring all metrics are bounded to $[0, 1]$ where higher values indicate better clustering quality. Each configuration is scored on the entire window (the per-point cluster assignments produced during LEARNONE are retained for this purpose), so no sub-sampling is required. The expected clustering quality for configuration $\theta$ at time $t$ is:

$$\mu_t(\theta) = \frac{1}{3}[\text{NSC} + \text{NDBI} + \text{NCHI}] \tag{2}$$

Equal weights avoid dataset-specific tuning and are consistent with the finding that no single CVI dominates across cluster shapes and densities [15], [16].

AutoDSC adjusts exploration based on the change in the best score between consecutive windows, $\Delta\mu_t = \mu_t(\theta_t^*) - \mu_{t-1}(\theta_{t-1}^*)$. The exploration rate then adapts as:

$$\alpha_{t+1} = \begin{cases} \max(0.1,\ \alpha_t \cdot (1 - \beta_t)) & \text{if } \Delta\mu_t > 0 \\ \min(1.0,\ \alpha_t \cdot (1 + \beta_t)) & \text{if } \Delta\mu_t \le 0 \end{cases} \qquad \beta_t = \min(\gamma, |r_t|) \tag{3}$$

where $r_t = \Delta\mu_t / |\mu_{t-1}(\theta_{t-1}^*)|$ is the relative improvement and $\beta_t$ is the resulting step, capped at $\gamma = 0.5$ for stability. This parameter-free mechanism decreases exploration when performance improves (favoring exploitation) and increases it when performance stagnates or drops, as happens under concept drift, encouraging broader search to recover.

The framework employs **PGSSO** (Algorithm 2) to jointly optimise algorithm selection and hyperparameter tuning. PGSSO maintains an elite archive $\mathcal{E}_t = \{(\theta_i, s_i)\}_{i=1}^{e}$ where each configuration $\theta_i = (a_i, \phi_i)$ pairs an algorithm choice with its hyperparameters and $s_i = \mu_t(\theta_i)$ is its score. To remain responsive to drift, elites are ranked by a *recency-weighted* score $s_i/(1+\Delta_i)$, where $\Delta_i$ is the number of windows since $\theta_i$ was last scored, so stale high-scoring configurations are down-weighted in favour of recently effective ones. By tracking which algorithm-hyperparameter combinations perform best, PGSSO naturally learns both which algorithms suit current data characteristics and their optimal settings. The elite population encodes this joint preference, where frequently appearing algorithms indicate superior algorithmic choices, while their parameter distributions reveal effective settings. PGSSO adaptively balances exploration and exploitation through dynamic rate $\alpha_t$, computing statistical distributions over both algorithm selections (via frequencies) and their parameters (via means $m_j$ and variances $\sigma_j^2$), which together guide sampling of new promising algorithm-hyperparameter configurations.

PGSSO generates new configurations through probabilistic sampling:

$$\theta_{\text{new}} = \begin{cases} \text{Elite-Guided}(\mathcal{E}_t) & \text{with probability } (1 - \alpha_t) \\ \text{Random}(\Theta) & \text{with probability } \alpha_t \end{cases} \tag{4}$$

Elite-guided sampling first draws an algorithm with probability equal to its frequency in the elite archive, then samples that algorithm's numerical hyperparameters from a Gaussian fitted to the elites using it:

$$\phi_j \sim \mathcal{N}(m_j, \sigma_j^2) \tag{5}$$

where the mean $m_j$ and variance $\sigma_j^2$ are computed over the elites that use the selected algorithm. Samples outside a

hyperparameter's defined bounds are clipped to the boundary. The elite-distribution sampling is related to Estimation of Distribution Algorithms and CMA-ES [21], adapted here for the non-stationary streaming setting.

When the population exceeds size $K$, it is pruned by a *farthest-first* rule: keep the incumbent and the top-scoring configurations, then greedily add whichever remaining configuration is most distant from those already kept. The distance between two configurations is 1 if their algorithms differ and the mean normalised per-hyperparameter distance otherwise, preserving diversity without a fixed acceptance threshold.

Throughout, AutoDSC emits a clustering for each instance using the incumbent $\theta_t^*$, the highest-scoring configuration in the current population.

All remaining constants are held fixed across every experiment. The exploration rate is clipped to $[0.1, 1.0]$ with a per-step cap $\gamma = 0.5$, the elite archive holds $e_{\max} = \lfloor K/2 \rfloor$ configurations, and each window PGSSO proposes $n_{\text{gen}} = \max(3, \lfloor K/4 \rfloor)$ new candidates (three at the default $K = 10$) that replace the lowest-ranked members so the population size stays at $K$. The Calinski–Harabasz score is rescaled by a fixed factor of 10 to bound NCHI to $[0, 1]$. The sensitivity study (Section VI-B) varies only $K$ and $w$ and holds all other constants at these defaults.

### A. Computational Complexity

The framework's computational complexity scales efficiently with population size and window parameters. Per-instance processing requires $O(K \cdot C)$ operations where $K = |\mathcal{A}_t|$ and $C$ denotes average algorithm complexity. Window-end evaluation involves $O(K \cdot w \cdot d)$ operations to score each configuration on the full window. Total per-window complexity is $O(K \cdot w \cdot d + K^2)$, where $K^2$ accounts for diversity-based population management. Crucially, $K$ is fixed, so per-point cost does not grow with stream length. Space complexity is $O((K + e_{\max}) \cdot p + w \cdot d)$, where $p$ is the number of parameters stored per configuration.

**Algorithm 1:** AutoDSC

**Input:** Stream $\mathcal{S} = \{x_1, x_2, \ldots\}$, $x_t \in \mathbb{R}^d$; window size $w$; population size $K$
**Output:** Cluster predictions $\{y_t\}_{t=1}^{\infty}$ using $\theta_t^*$

$e_{\max} \leftarrow \lfloor K/2 \rfloor$;
$\mathcal{E}_0 \leftarrow \emptyset$, $\alpha_0 \leftarrow 0.5$;
$\mathcal{A}_0 \leftarrow \{\theta_i\}_{i=1}^{K}$ where $\theta_i \sim \text{Uniform}(\Theta)$;
$W_0 \leftarrow \emptyset$, $\theta_0^* \leftarrow \emptyset$;
$\mu_{-1}(\theta_{-1}^*) \leftarrow -\infty$;
**foreach** $x_t \in \mathcal{S}$ **do**
  $W_t \leftarrow W_{t-1} \cup \{x_t\}$;
  **foreach** $\theta_i \in \mathcal{A}_t$ **do**
    $\ell_{i,t} \leftarrow \text{PREDICTONE}(M_i, x_t)$ ; `// store assignment`
    $M_i \leftarrow \text{LEARNONE}(M_i, x_t)$;
  **if** $|W_t| = w$ **then**
    **foreach** $\theta_i \in \mathcal{A}_t$ **do**
      $L_i \leftarrow \{\ell_{i,\tau} : x_\tau \in W_t\}$;
      $\mu_t(\theta_i) \leftarrow \text{NORMCVI}(W_t, L_i)$;
    $\mu_{\text{best}}^{(t)} \leftarrow \max_i \mu_t(\theta_i)$;
    $\alpha_{t+1} \leftarrow \text{ADAPT}(\alpha_t, \mu_{t-1}(\theta_{t-1}^*), \mu_{\text{best}}^{(t)})$;
    $\mathcal{C}_{\text{new}} \leftarrow \text{PGSSO}(\{(\theta_i, \mu_t(\theta_i))\}, \alpha_{t+1}, \Theta)$;
    $\mathcal{A}_{t+1} \leftarrow \text{SELECTDIVERSE}(\mathcal{A}_t \cup \mathcal{C}_{\text{new}}, K)$;
    $\theta_t^* \leftarrow \arg\max_{\theta_i \in \mathcal{A}_{t+1}} \mu_t(\theta_i)$;
    $W_t \leftarrow \emptyset$;

**Algorithm 2:** PGSSO

**Input:** Current configurations with scores $\{(\theta_i, s_i)\}$; exploration rate $\alpha \in [0, 1]$; parameter space $\Theta$
**Output:** New configuration set $\mathcal{C}_{\text{new}}$

$\mathcal{E} \leftarrow \text{UPDATEELITEPOPULATION}(\{(\theta_i, s_i)\}, e_{\max})$; `// recency-weighted ranking, see §IV`
$\mathcal{E} \leftarrow \text{UPDATEPARAMETERDISTRIBUTIONS}(\mathcal{E})$; `// fit` $\mathcal{N}(m_j, \sigma_j^2)$ `per algorithm, see §IV`
$\mathcal{C}_{\text{new}} \leftarrow \emptyset$;
**for** $j \leftarrow 1$ **to** $n_{gen}$ **do**
  $u \leftarrow \text{Uniform}(0, 1)$;
  **if** $u > \alpha$ **then**
    $\theta_{\text{new}} \leftarrow \text{SAMPLEFROMELITE}(\mathcal{E}, \Theta)$;
  **else**
    $\theta_{\text{new}} \leftarrow \text{RANDOMSAMPLE}(\Theta)$;
  $\mathcal{C}_{\text{new}} \leftarrow \mathcal{C}_{\text{new}} \cup \{\theta_{\text{new}}\}$;
**return** $\mathcal{C}_{\text{new}}$;

## V. Experimental Setup

Following standard AutoML benchmarking practice [22], our experimental design uses rigorous, reproducible evaluation protocols to ensure statistically valid results.[1]

### A. Datasets

We evaluate on 10 datasets comprising 5 real-world and 5 synthetic streams with varying sizes, dimensionalities, and cluster counts, as summarized in Table I. Datasets were selected to (i) cover the real-world streams most frequently used in prior stream-clustering work, spanning a wide range of sizes (11k–101k instances), dimensionalities (9–100 attributes), and cluster counts (3–12), and (ii) probe specific drift mechanisms in isolation through controlled synthetic generators.

The five real-world streams are public benchmarks with natural class structure and distribution change.

- **Insects** [23]: optical-sensor wing-beat features of flying insects, where the species labels and feature distribution shift with ambient temperature over the recording.
- **KDD99** [24]: network-connection records from the KDD Cup 1999 intrusion-detection task, labelled as normal traffic or one of several attack types.

[1]Datasets and code: https://anonymous.4open.science/r/AutoDSC

TABLE I
SUMMARY OF DATASETS. FIVE REAL-WORLD AND FIVE SYNTHETIC STREAMS SPANNING A RANGE OF SIZES, DIMENSIONALITIES, AND CLUSTER COUNTS.

| Dataset Name | Type | Instances | Attributes | Clusters |
|---|---|---|---|---|
| Insects | Real | 52,848 | 33 | 6 |
| KDD99 | Real | 100,655 | 38 | 12 |
| PenDigits | Real | 10,992 | 16 | 10 |
| Shuttle | Real | 58,000 | 9 | 7 |
| Vehicle | Real | 98,528 | 100 | 3 |
| Expanding | Synth | 100,000 | 5 | 5 |
| Gradual | Synth | 100,000 | 5 | 5 |
| Merge-Split | Synth | 100,000 | 5 | 5 |
| Sudden | Synth | 100,000 | 5 | 5 |
| Fast-RBF | Synth | 100,000 | 5 | 5 |

- **PenDigits** [24]: pen-trajectory features of handwritten digits, with one cluster per digit 0–9.
- **Shuttle** [24]: NASA Space Shuttle radiator sensor readings (Statlog), with highly imbalanced operational-state classes.
- **Vehicle** [25]: high-dimensional acoustic and seismic features from a distributed wireless sensor network, labelled by vehicle type (note: distinct from the unrelated UCI Vehicle Silhouettes dataset).

The five synthetic streams are produced by our reproducible generators (released with the code) and let us control the drift mechanism while holding dimensionality and cluster count fixed. Four are five-cluster Gaussian-blob streams in $[0, 1]^5$ that differ only in the injected drift, and one is a MOA-style RandomRBF stream [26].

- **Expanding**: cluster centroids stay fixed while the within-cluster variance grows steadily over the stream.
- **Gradual**: centroids translate smoothly at constant speed along fixed random directions.
- **Merge-Split**: clusters converge toward the global mean at mid-stream and then re-separate.
- **Sudden**: centroids jump abruptly at three change points, holding stationary between them.
- **Fast-RBF**: a RandomRBF stream whose class centroids drift rapidly across the feature space.

### B. Evaluation Protocol

All experiments use the prequential protocol (Section II) with 10 independent runs using random seeds 1-10 to account for stochastic variability. Clustering quality is measured using ARI and CMM [17], where ground truth labels are used solely for evaluation, not training or model selection. ARI is the chance-corrected form [27] of the Rand index [28] that counts agreements between the predicted clustering and the ground-truth partition over all pairs of points and adjusts the count for the agreement expected at random, yielding 1 for a perfect match, 0 for random labelling, and negative values for worse-than-random partitions. We also record throughput (instances/s) and peak memory usage (MB) to assess computational efficiency. Statistical significance is assessed using the Friedman test followed by the Nemenyi post-hoc test [29] with $\alpha = 0.05$.

TABLE II
SPECIFICATION OF OUR EVALUATION PLATFORM.

| Component | Description |
|---|---|
| Processor | 2× AMD EPYC 7702 64-Core @ 2.0 GHz (128 cores) |
| L3 Cache | 512 MiB |
| Memory | 1 TiB DDR4 RAM |
| OS | Ubuntu 24.04.4 LTS |
| Kernel | Linux 6.8.0-101-generic |
| Runtime | Python 3.12.3, River [6] |
| Compiler | GCC 13.3.0 |

### C. Baseline Methods

We compare AutoDSC against five baselines spanning three categories. (i) *AutoML stream-clustering methods*: *ConfStream* [7] and *OACE* [8], established frameworks that optimise hyperparameters online. (ii) The *rule-based adaptive* method *MOStream* [9]. (iii) Two *baselines*: *RandomSearch* [30], which shares AutoDSC's framework structure but generates configurations by purely random sampling, a direct comparison to our guided optimisation, and the *Default(Best)* baseline of OACE [8], which captures that "one of the most commonly-used approaches by non-expert users is to try all possible clustering algorithms with their default hyperparameters and then select the model with the best performance." Following this, we run the four River fixed algorithms (CluStream, DenStream, DBSTREAM, and StreamKMeans) with their default settings and report the best-performing model per dataset.

### D. Implementation and Configuration

Unless otherwise stated, AutoDSC uses its default configuration of population size $K = 10$ and window size $w = 1000$, with the exploration rate adapting automatically from an initial value of 0.5 (Section VI-B analyses sensitivity to $K$ and $w$). The AutoML baselines search over the same algorithm portfolio and hyperparameter ranges specified in Table III, where the window-dependent ranges (`time_window` and `chunk_size`) are listed for the default window size $w = 1000$; the single fixed algorithms use their library defaults. To ensure a fair comparison, all AutoML baselines receive the same window size and per-window evaluation budget as AutoDSC, so differences reflect the search and selection strategy rather than the compute allotment; other baseline-specific parameters remain as reported in their original papers.

All experiments were conducted on the server detailed in Table II, using parallelisation across cores to run the ten independent seeds concurrently; all methods process the stream single-threaded within each run, so throughput comparisons are fair. All clustering algorithms were implemented with the River online-learning library [6].

## VI. RESULTS

Table IV presents a comprehensive comparison of AutoDSC against five baselines across ten diverse real-world

TABLE III
SEARCH SPACE CONFIGURATION. PER-ALGORITHM HYPERPARAMETER TYPES AND RANGES SEARCHED BY AUTODSC AND THE AUTOML BASELINES.

| Algorithm | Hyperparameter | Type | Range/Options |
|---|---|---|---|
| CluStream | n_macro_clusters | integer | [2, 15] |
| | max_micro_clusters | integer | [20, 100] |
| | time_window | integer | [250, 1500] |
| | micro_cluster_r_factor | numeric | [1.0, 3.0] |
| DenStream | decaying_factor | numeric | [0.01, 0.2] |
| | beta | numeric | [0.2, 0.8] |
| | mu | numeric | [2.0, 8.0] |
| | epsilon | numeric | [0.01, 1.0] |
| DBSTREAM | clustering_threshold | numeric | [0.1, 2.0] |
| | fading_factor | numeric | [0.001, 0.05] |
| | intersection_factor | numeric | [0.1, 0.8] |
| | minimum_weight | numeric | [0.5, 5.0] |
| StreamKMeans | chunk_size | integer | [100, 500] |
| | n_clusters | integer | [2, 20] |
| | halflife | numeric | [0.05, 0.8] |

and synthetic datasets, evaluating clustering quality with both ARI and CMM. Following OACE's *default-baseline* [8], we summarize the single fixed algorithms by trying all four River algorithms at their default hyperparameters and reporting the best-performing model per dataset (denoted Default(Best)). AutoDSC instead selects and tunes configurations automatically.

AutoDSC achieves the highest overall mean ARI of **0.622**, ahead of every baseline, surpassing the strongest AutoML competitor ConfStream (0.572) by 5.0 points, OACE (0.558) by 6.4 points, RandomSearch and MOStream (both 0.534) by 8.8 points, and the Default(Best) baseline (0.517) by 10.5 points, and it attains the best mean rank on both ARI and CMM. AutoDSC ranks first on ARI for 7 of the 10 datasets, with the largest margins on the drifting synthetic streams (e.g. Fast-RBF, Gradual, Expanding all $\geq 0.99$, where even the best default reaches only $0.76$–$0.82$) and on the harder real stream PenDigits ($0.510$ vs. $0.412$ for the best default). It is outranked on Shuttle and Vehicle, where a single default algorithm is already near-optimal, and on Merge-Split, where OACE leads narrowly ($0.985$ vs. $0.963$). Figure 2 shows the prequential trajectories, where AutoDSC stays at or above all baselines throughout the stream on both a real (PenDigits) and a synthetic (Expanding) workload, reflecting rapid adaptation as the stream evolves.

Figure 3 isolates AutoDSC's behaviour under drift on two synthetic streams. On *Merge-Split* (a), which contains a discrete merge-then-split event mid-stream (shaded), every adaptive method incurs a temporary drop in ARI and recovers within roughly fifteen windows; AutoDSC sustains the highest stable quality (ARI $\approx 1.0$) on either side of the drift and recovers together with the strongest baselines, consistent with its adaptive-exploration design. On *Expanding* (b), where within-cluster variance grows continuously throughout the stream (gradient), AutoDSC holds near-perfect ARI for almost the entire stream and degrades the least as the drift intensifies, whereas ConfStream tracks it early but destabilises sharply late in the stream, and OACE and MOStream stay well below it and steadily decline. AutoDSC thus both recovers from abrupt drift and best sustains quality under continuous drift through its adaptive exploration alone.

To assess these performance gains statistically, the Critical Difference (CD) diagrams in Figure 4 rank all methods using the Friedman test with the Nemenyi post-hoc procedure ($\alpha = 0.05$, CD$= 2.38$). AutoDSC attains the best (lowest) mean rank on both ARI ($1.40$) and CMM ($2.20$), ahead of every baseline. On ARI, AutoDSC is significantly better than RandomSearch, MOStream, and the Default(Best) baseline; the rank gap to the strongest AutoML competitors (ConfStream and OACE, both at rank $3.60$) is $2.20$, just within the critical difference of $2.38$, so with only ten datasets these are not separated by a statistically significant margin. On CMM, AutoDSC (rank $2.20$) is significantly better than MOStream and Default(Best), whose ranks fall beyond the CD boundary; the gap to ConfStream and OACE is within the CD, consistent with the ARI result. AutoDSC nonetheless ranks first on both metrics.

Beyond clustering quality, AutoDSC remains practical in throughput and memory. Among the AutoML methods it is the fastest, processing **2.8× more instances/s** than ConfStream and **2.0× more** than OACE, while using less memory than both (≈25 MB vs. 27 and 53 MB). The Default(Best) baseline is naturally faster as it performs no search, and RandomSearch trades quality for speed; AutoDSC instead occupies the favorable region of the throughput–memory plane (Figure 5), pairing the highest clustering quality with a low memory footprint. OACE shows the worst AutoML footprint (53 MB) and MOStream the highest overall (166 MB).

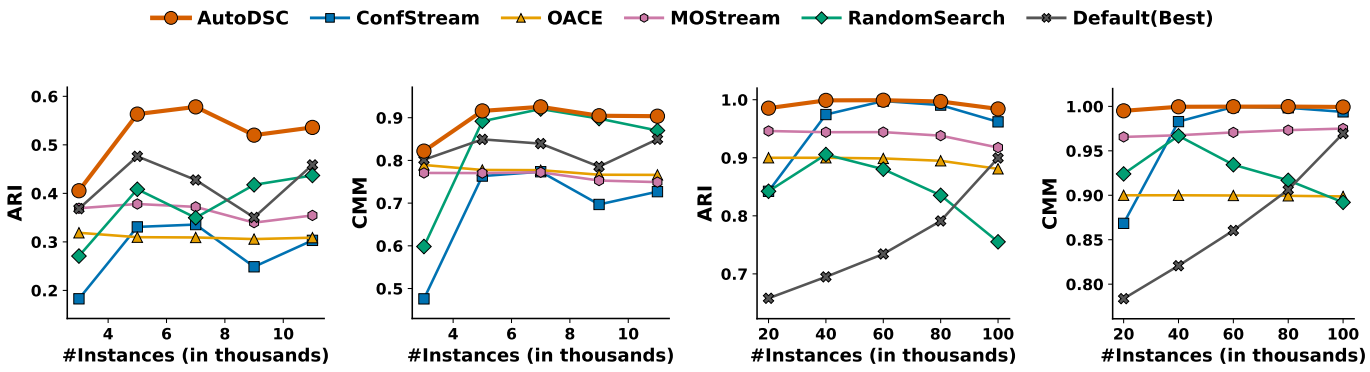
To quantify the contribution of each component, we ran a leave-one-out ablation across datasets ($\times 10$ seeds), disabling one mechanism at a time, namely recency-weighted elite ranking, the adaptive exploration rate, the multi-objective (3-CVI) score, and diversity-preserving pruning. (Replacing the guided PGSSO search by random configuration generation is already covered by the RandomSearch baseline.) Figure 6 reports each variant's score relative to the full model. On these comparatively stable real-world streams the effects are small and metric-dependent: removing recency weighting causes the largest ARI drop ($\sim$1.7%), while removing the multi-objective CVI score lowers CMM by $\sim$1.2% but barely affects ARI, indicating that the two components matter on complementary metrics; diversity pruning gives a small consistent gain, and the adaptive exploration rate has a negligible standalone effect here, consistent with its role being to accelerate recovery under the stronger drift of the synthetic streams (Figure 3). The full configuration is the most balanced across both metrics, supporting the integrated design.

Finally, AutoDSC adaptively reshapes its population of candidate algorithms as the stream evolves. Figure 7 shows the algorithm composition of the population across stream windows on the Insects dataset, chosen as a representative real-world stream: it is the longest real stream (52,848 instances) and exhibits pronounced concept drift as species-label distributions shift with ambient temperature, where each cell is the

TABLE IV

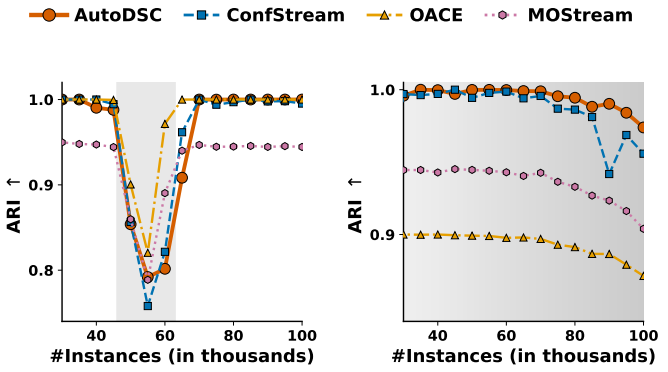
CLUSTERING QUALITY ACROSS DATASETS. ARI AND CMM (↑ HIGHER IS BETTER) OVER 10 SEEDS; PER-DATASET CELLS REPORT MEAN±STD AND THE MEAN SCORE ROW REPORTS THE MEAN ACROSS DATASETS. BEST PER METRIC PER ROW IN **BOLD**, SECOND BEST UNDERLINED.

| Dataset | AutoML | | | | | | Rule-based | | Baselines | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | AutoDSC | | ConfStream | | OACE | | MOStream | | RandomSearch | | Default(Best) | |
| | ARI | CMM | ARI | CMM | ARI | CMM | ARI | CMM | ARI | CMM | ARI | CMM |
| Insects | **.117±.003** | .905±.024 | .016±.004 | .827±.019 | .048±.035 | **.969±.028** | .090±.045 | .964±.006 | .054±.007 | .880±.019 | .100±.002 | .947±.001 |
| KDD99 | **.037±.000** | .942±.015 | .037±.001 | .910±.013 | .036±.003 | .626±.133 | .018±.014 | .780±.174 | .033±.001 | .935±.017 | .037±.000 | **.990±.000** |
| PenDigits | **.510±.012** | **.888±.014** | .271±.029 | .668±.025 | .311±.120 | .777±.107 | .363±.010 | .764±.016 | .367±.035 | .814±.016 | .412±.000 | .823±.000 |
| Shuttle | .363±.011 | .900±.006 | .341±.002 | .887±.000 | .146±.083 | **.974±.034** | .002±.001 | .380±.054 | .270±.014 | .903±.018 | **.409±.000** | .886±.000 |
| Vehicle | .320±.002 | .975±.005 | .303±.001 | .967±.000 | .312±.003 | .976±.000 | .335±.017 | **.990±.001** | .172±.017 | .977±.001 | **.335±.003** | .986±.000 |
| Expanding | **.993±.001** | **.999±.001** | .953±.028 | .969±.020 | .895±.298 | .900±.300 | .938±.111 | .970±.058 | .844±.017 | .927±.011 | .756±.103 | .868±.055 |
| Gradual | **.998±.001** | 1.000±.000 | .956±.042 | .964±.032 | .995±.014 | **1.000±.000** | .978±.066 | .984±.047 | .921±.014 | .978±.004 | .759±.130 | .838±.094 |
| Merge-Split | .963±.004 | .988±.001 | .952±.009 | .979±.007 | **.985±.001** | **.999±.000** | .932±.102 | .968±.062 | .877±.012 | .947±.009 | .895±.054 | .945±.034 |
| Sudden | **.921±.001** | **.956±.001** | .904±.013 | .939±.010 | .855±.003 | .884±.000 | .774±.090 | .841±.067 | .878±.006 | .947±.002 | .646±.160 | .757±.116 |
| Fast-RBF | **.999±.001** | **1.000±.000** | .988±.001 | .989±.000 | .995±.008 | .999±.002 | .911±.109 | .938±.076 | .924±.009 | .963±.005 | .818±.096 | .896±.056 |
| **Mean Score** | **.622** | **.955** | .572 | .910 | .558 | .910 | .534 | .858 | .534 | .927 | .517 | .894 |
| **Mean Rank** | **1.40** | **2.20** | 3.60 | 4.30 | 3.60 | 2.90 | 4.10 | 3.80 | 4.50 | 3.50 | 3.80 | 4.30 |



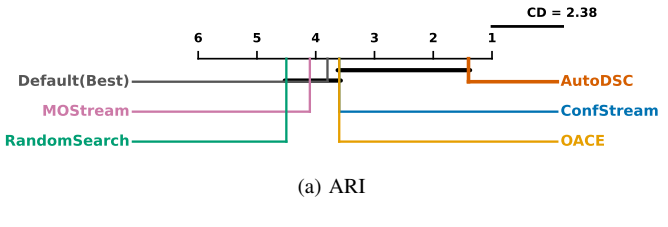

(a) PenDigits ARI (b) PenDigits CMM (c) Expanding ARI (d) Expanding CMM

Fig. 2. Prequential Clustering Quality Across Stream Stages. Results for all methods on a real (PenDigits) and a synthetic drifting (Expanding) stream; AutoDSC is drawn thick.

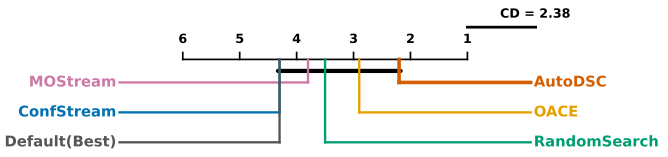



(a) *Merge-Split* (abrupt) (b) *Expanding* (continuous)

Fig. 3. Drift Behaviour on Two Synthetic Streams. Per-window ARI (seed-averaged); AutoDSC sustains the highest quality and recovers under both an abrupt event (a) and continuous drift (b), with the y-axes zoomed to the post-warm-up range.




(a) ARI




(b) CMM

Fig. 4. Critical Difference Diagrams for ARI and CMM. Ranks from the Friedman test with the Nemenyi post-hoc procedure ($\alpha = 0.05$); methods connected by a bar are not significantly different.

base algorithm occupying a population slot at that window. AutoDSC continuously maintains a diverse mix and shifts the composition between CluStream, DenStream, DBSTREAM, and StreamKMeans as the active concept changes, rather than committing to a single algorithm.

### A. Supplementary Analyses

**Label-free selection is sound.** AutoDSC selects the incumbent each window with a label-free internal criterion, so its value hinges on that criterion tracking the unobserved external quality. On a real (PenDigits) and a drifting synthetic (Sudden) stream we bucket each window into terciles of the incumbent's internal multi-CVI score and measure the mean external ARI

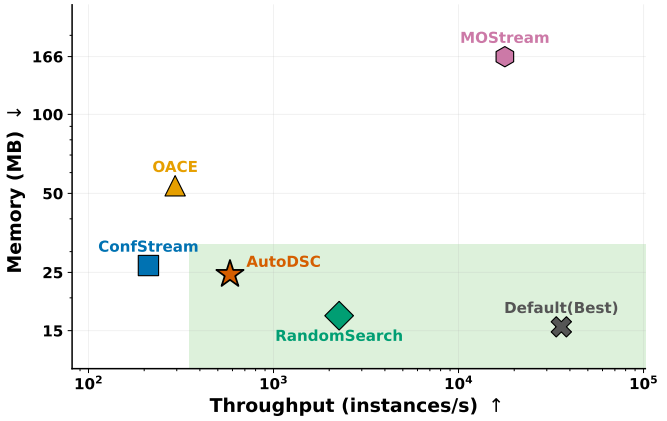



Fig. 5. Throughput–Memory Trade-off. Mean over all datasets on log–log axes; the shaded region marks the favorable high-throughput, low-memory area, where AutoDSC (⋆) sits.

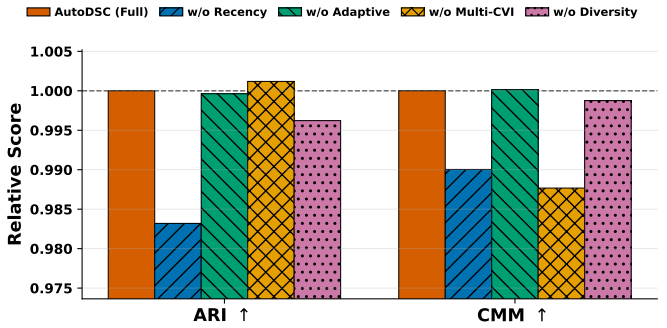



Fig. 6. Leave-One-Out Ablation on Real-world Streams. ARI and CMM of each variant relative to the full AutoDSC (dashed line = 1.0).

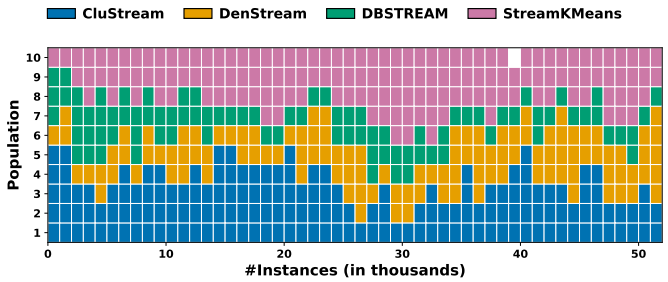
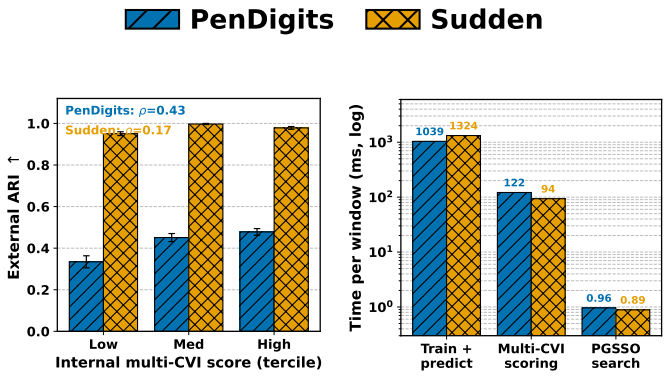



Fig. 7. Population Composition over the Insects Stream. Colors denote the selected base algorithm per population slot (rows) and stream stage (columns).




(a) Label-free selection validity (b) Per-window runtime breakdown

Fig. 8. Supplementary Analyses on PenDigits and Sudden. Mean over 10 seeds; AutoDSC never observes labels or the cluster count $k$. (a) External ARI rises with the incumbent's internal multi-CVI score (Spearman $\rho$ shown), so label-free selection is sound. (b) Per-window runtime (log scale): the PGSSO search is three orders of magnitude below base-clusterer training, i.e. negligible.

per bucket (Figure 8a). On PenDigits, ARI rises monotonically with the internal score, from 0.33 in the lowest tercile to 0.48 in the highest (Spearman $\rho = 0.43$); on Sudden nearly every window already attains ARI $\approx 1.0$, leaving little headroom to separate ($\rho = 0.17$). The label-free score is thus a sound, if imperfect, basis for selection, consistent with the limitation noted in Section VIII. More strikingly, although AutoDSC never observes the number of clusters, the incumbent's mean per-window cluster count matches the ground truth almost exactly (10.2 vs. 10 on PenDigits, 5.0 vs. 5 on Sudden), addressing the long-standing problem of choosing $k$ online.

**The search overhead is negligible.** Decomposing AutoDSC's per-window wall-clock into base-clusterer training/prediction, multi-CVI scoring, and the PGSSO search itself (Figure 8b) reveals three well-separated tiers: training the $K$ population members dominates ($\approx$ 1.0–1.3 s/window), multi-CVI scoring is an order of magnitude smaller ($\approx 0.1$ s), and the AutoML search machinery is three orders smaller still ($\approx 1$ ms, under 0.1% of the total). The price of automation is thus dwarfed by the clustering it automates, explaining AutoDSC's competitive throughput against the offline optimisation of OACE.

**Quality is stable across dimensionality.** On a controlled five-cluster blob stream with feature dimensionality growing from 5 to 80 (Figure 9), clustering quality stays high and even improves slightly (ARI 0.91 $\rightarrow$ 0.97) while throughput degrades only gracefully (620 $\rightarrow$ 376 instances/s), with no curse-of-dimensionality collapse, indicating that AutoDSC's search remains effective as dimensionality increases.

### B. *Sensitivity Analysis*

We analyse AutoDSC's sensitivity to its two free hyperparameters, window size $w$ and population size $K$, averaged over all datasets ($\times$10 seeds), jointly reporting clustering quality (ARI) and throughput (Figure 10). Window size shows a clear quality optimum at $w = 1000$, where smaller windows give insufficient data for reliable per-window CVI evaluation, while larger windows delay adaptation to drift and reduce throughput. Population size exhibits a quality plateau, where ARI rises steeply from $K = 5$ and saturates by $K \approx 15$, while throughput falls monotonically as $K$ grows. The knee of this trade-off is at $K = 10$, which attains near-peak quality at substantially higher throughput, justifying it as the default. Overall, AutoDSC is stable within reasonable ranges and degrades gracefully outside them, with the default configuration

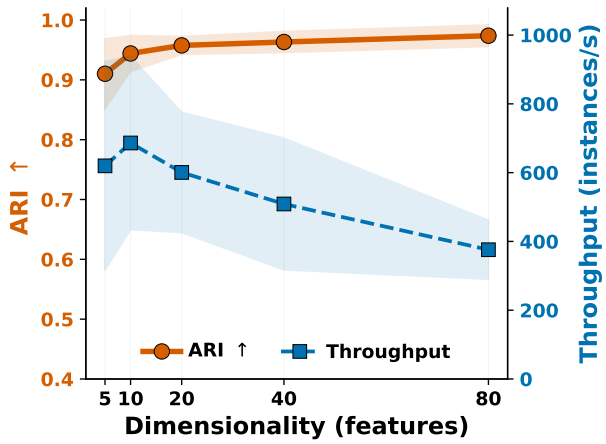
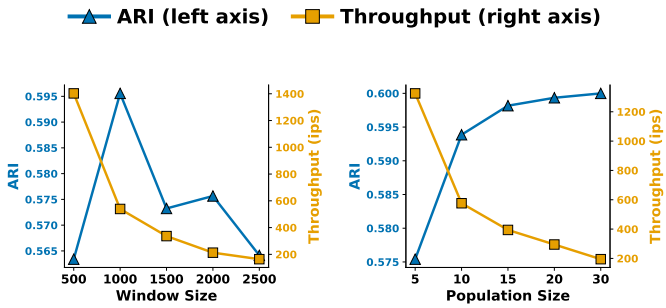



Fig. 9. Dimensionality Scalability of AutoDSC. Controlled five-cluster blob stream (mean over 10 seeds, shaded ±std); clustering quality (ARI) stays stable while throughput degrades only gracefully as the feature count grows.




(a) Window Size (b) Population Size

Fig. 10. Sensitivity to Window Size and Population Size. Averaged over all datasets.

($w = 1000$, $K = 10$) sitting at the quality/efficiency sweet spot of both parameters.

## VII. Prior Work

AutoML has been brought to batch clustering—AutoML4Clust, AutoCluster, cSmartML, ML2DAC, TPE-AutoClust, and PoAC [31]–[36]—and to streaming *supervised* tasks [37]–[40], but neither line operates under the single-pass, label-free, drifting conditions of stream clustering, for which the state of the art is three systems that differ in their search strategy, selection criterion, and main limitation. *ConfStream* [7] maintains an ensemble of configurations and resamples near the better-performing ones, selecting by a single Silhouette-Width score that biases the search toward one cluster shape and proves brittle across datasets. *OACE* [8] searches with Tree-structured Parzen Estimator Bayesian optimisation over clustering ensembles and selects by an internal ensemble vote, but depends on a computationally heavy offline tuning budget that leaves it sensitive to its time allowance and slow to recover under rapid drift. *MOStream* [9], the automated successor of the modular Benne algorithm from the SESAME study [41], reconfigures four structural design dimensions by rule rather than searching the joint algorithm–hyperparameter space, so it self-optimises structurally but never jointly tunes continuous hyperparameters with the algorithm choice. AutoDSC departs on all three axes: a single-pass, population-guided search over the *joint* algorithm–hyperparameter space (vs. OACE's offline optimisation and MOStream's structural rules), selection by a *multi-objective* label-free criterion (vs. ConfStream's single metric), and implicit drift adaptation through continual per-window re-evaluation.

## VIII. Discussion and Limitations

AutoDSC advances AutoML for streaming environments through several design choices. Its multi-objective criterion, combining three normalised cluster-validity indices, prevents convergence to configurations that excel on one aspect while failing catastrophically on others [15], [16], and PGSSO's adaptive exploration rate enables rapid recovery after concept drift while exploiting good configurations during stable periods. We emphasise that AutoDSC's clearest advantages are its full autonomy, requiring no manual tuning or labels, and its efficiency, together with statistically significant gains over untuned single algorithms and random search; we therefore frame its value as delivering competitive quality fully automatically and at lower cost rather than as a uniform quality win (scope and significance caveats are detailed below). These contributions bridge traditional AutoML and the dynamic requirements of real-time streaming applications in IoT, sensor networks, and analytics platforms, where manual configuration is impractical [2].

Several limitations bound the scope of our claims. First, with only ten datasets the quality advantage of AutoDSC over the strongest AutoML competitors (ConfStream and OACE) is not statistically significant, although AutoDSC attains the best mean rank and significantly outperforms untuned single algorithms and random search. Second, our evaluation targets tabular streams with numerical and categorical features, so extension to sparse text, multi-modal, or image streams would require domain-specific base clusterers. Third, AutoDSC still exposes two free hyperparameters, the window size and the population size, which require sensible defaults, although it remains stable across a broad range of both. Fourth, while PGSSO supports categorical hyperparameter sampling, the current portfolio uses only continuous and integer parameters; algorithms with categorical choices (e.g. distance metric) would exercise this capability directly. Finally, configuration selection relies on internal validity indices computed without labels, so online clustering quality is estimated rather than observed, and a misleading internal score can temporarily mislead selection between ground-truth evaluations.

## IX. Conclusion and Future Work

We presented AutoDSC, a single-pass AutoML framework for data stream clustering that proposes configurations with a population-guided search over a recency-weighted elite archive and selects among them with a label-free multi-objective criterion, adapting implicitly as the stream drifts. Across ten

streams it attains the best mean rank on both ARI and CMM and the highest mean ARI of all methods—significantly better than untuned single algorithms—while being the fastest and most memory-efficient AutoML stream clusterer. Future work includes semi-supervised and multi-task extensions and richer multi-objective optimisation for dynamic settings.

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